

Nguyen Anh Duc

HMI / Embedded / IoT Engineer

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Summary

HMI / Embedded Systems Engineer with 1+ year of hands-on experience developing real-time instrumentation, data acquisition, and hardware-integrated software. Experienced in C# WPF, Python/PyQt6, high-speed binary serial communication, real-time visualization, and hardware/software integration. Background in embedded connectivity including BLE GATT, LTE-M/NB-IoT, and UART. Currently on international secondment as Junior Firmware Developer with THESIS PTE LTD (Singapore), expanding into ARM Cortex-M firmware development.

Experience

THESIS PTE LTD, Junior Firmware Developer (Secondment) – Singapore Sept 2026 – present

- Developing and maintaining embedded firmware in C/C++ for ARM Cortex-M platforms under senior engineer guidance.
- Working with peripheral interfaces including UART, SPI, I²C, GPIO, and ADC across bare-metal and RTOS-based systems.
- Supporting hardware bring-up, firmware/PCB integration, and debugging using J-Link, SWD, and serial diagnostics.
- Investigating firmware issues, participating in code reviews, and maintaining technical documentation.

Enable Startup, HMI / Embedded / IoT Engineer – Da Nang, Vietnam June 2025 – present

- Architected a multi-process scientific control workbench in Python (PyQt6, PyQtGraph) for precision optical metrology; decoupled FPGA data acquisition into a dedicated OS process via IPC queue, implemented phase-sensitive demodulation with 4th-order SOS Butterworth filtering, and designed a closed-loop auto-lock state machine achieving sub-millivolt zero-crossing convergence while maintaining 60 FPS OpenGL rendering.
- Designed and implemented high-performance desktop HMI applications in C# .NET 8 WPF (MVVM + ScottPlot 5) for multi-channel scientific sensor acquisition over 1 Mbps binary UART links; engineered pre-allocated flat buffer pipelines sustaining 200,000-point real-time visualization at 30 FPS during 48-hour continuous soak testing.
- Implemented a 25-channel binary serial DAQ system with a circular ring buffer parser resolving byte-boundary misalignment at 100 Hz, paired with an automated 5-step in-situ polynomial calibration wizard.
- Developed application-level firmware for a PSoC6 host controller interfacing an nRF9160 LTE-M/NB-IoT modem via AT commands over UART; diagnosed UART buffer desynchronization and pin-mapping conflicts through serial-log inspection and oscilloscope signal tracing.
- Designed Python host application integrating a 3-axis CNC motion controller via native C-DLL (ctypes) with explicit struct alignment; engineered continuous fly-by rastering achieving 7× scan speedup (500 × 500 volumetric grid in < 22 minutes).
- Implemented a concurrent dual-port serial ATE system (921,600 baud and 1,000,000 baud simultaneously) with WMI-based COM port auto-discovery; reduced PCB factory functional test cycle time from 10 minutes to under 45 seconds.
- Collaborated with hardware engineers and Singapore-based research partners on system integration, hardware-software validation, and technical requirements translation.

Skills

Embedded & Firmware: C, C++, ARM Cortex-M, Nordic nRF, STM32, PSoC 6 / 5LP, ESP32, Bare-metal, FreeRTOS, UART, SPI, I²C, GPIO, ADC

Communication & Protocols: BLE GATT Client, LTE-M / NB-IoT (AT Commands), MQTT, LoRa / LoRaWAN, TCP/IP, SCPI over TCP, Binary Custom Framing, CRC16

HMI & Desktop Software: Python (PyQt6, PyQtGraph, SciPy, NumPy, ctypes), C# (.NET 8, WPF, MVVM), ScottPlot 5, Real-Time Visualization, Multiprocessing IPC, Producer-Consumer Pipelines

Debug & Test Equipment: J-Link (Segger), SWD / JTAG, Oscilloscope, Logic Analyzer (Saleae), Serial Port Monitor, Wireshark, dotMemory, Visual Studio Diagnostic Tools

Tools & Platforms: Git, Linux (Ubuntu, Raspberry Pi OS), CMake, VS Code, Visual Studio 2022, Android Studio, Kotlin 2.0

Projects

Real-Time Laser Frequency Stabilization Workbench

Scientific Instrumentation — FPGA / DSP / Closed-Loop Control

- Implemented a real-time 4th-order SOS Butterworth filtering and phase-sensitive demodulation pipeline with persistent filter state across streaming chunks; designed a deterministic auto-lock state machine with zero-crossing capture, 2-stage anti-windup, and actuator rail saturation detection.
- Achieved 60 FPS real-time 4-channel oscilloscope rendering and < 1 mV zero-crossing convergence window.
- Tools: PyQt6, PyQtGraph (OpenGL), SciPy, NumPy, Multiprocessing IPC, Digital PID.

Industrial BLE OTA & Field Diagnostic Tool

Embedded Mobile Tool — Kotlin / BLE GATT / Bootloader Protocol

- Ported the proprietary Cypress DFU protocol from legacy Python into an asynchronous Kotlin Coroutines engine on Android BLE GATT, enabling wireless field upgrades for PSoC 6 dual-core (.cyacd2) and PSoC 5LP (.cyacd) devices without transport laptops.
- Engineered binary container parsers with table-driven CRC-32C (Castagnoli, 0x82F63B78) calculation and an 8-stage DFU state machine; implemented dual transmission modes (Safe Mode with chunk-level ACKs vs. Fast Mode with unacknowledged writes and GATT queue-saturation back-off).
- Achieved 100% flash success rate across 150+ field test cycles under 2.4 GHz factory interference, while integrating real-time 3-axis vibration FFT/PSD spectrum streaming with ISO 10816 condition monitoring overlays at 60 FPS.
- Tools: Kotlin 2.0, Jetpack Compose, Android BLE GATT, Cypress DFU Protocol, Coroutines, StateFlow, PSoC 6 / 5LP.

Smart Water Meter & Edge AI Monitoring System

Graduation Thesis — Edge AI / IoT — Scored 8.7/10

- Designed an offline edge-AI pipeline for water meter digit recognition: ROI-based image preprocessing, INT8-quantized CNN inference on ESP32-CAM, and LoRa/MQTT uplink for remote consumption reporting.
- Achieved ~98% test accuracy on a self-collected dataset of 612 frames with 1,757 labeled digits.
- Tools: ESP32-CAM, TensorFlow Lite Micro, INT8 Quantization, LoRa, MQTT.

Education

Vietnam-Korea University of Information Technology and Communications (VKU),

Aug 2021 – Apr 2026

Bachelor of Science in Information Technology (Major: IoT & Robotics) – Da Nang, Vietnam

- GPA: 3.21 / 4.0 · Graduation Thesis: 8.7 / 10

Languages

English: VSTEP B2